

Introduction

Stroke is a leading cause of death and acquired disability worldwide. In 2021, the Global Burden of Disease Study estimated 11.9 million incident strokes and 7.3 million stroke deaths worldwide, with ischemic stroke accounting for 65.3% of incident cases (1). Survivors frequently retain neurological deficits carrying substantial personal, social, and economic costs, and the burden falls disproportionately on low-income and middle-income countries undergoing rapid demographic transition (1).

Acute stroke care has advanced with intravenous thrombolysis and endovascular thrombectomy (2), but these reach only a minority of patients because of narrow therapeutic windows, strict eligibility criteria, infrastructure requirements, and variable access (3). Many patients treated in time still have residual deficits, and recovery-oriented and preventive therapies are needed to address that gap (4). Traditional Chinese medicine is used alongside contemporary stroke care throughout East Asia, particularly in China, as single herbs, patent medicines, injections and multi-component prescriptions (5, 6).

Proposed mechanisms include neuroprotection and effects on post-ischemic neuroinflammation, oxidative stress, endothelial injury and cerebral microcirculation (7). Several formulations have been tested in adequately sized randomized trials, including Tongxinluo (8), Panax notoginseng saponins in the PANDA trial and its pre-specified secondary analysis (9, 10), Naioxintong (11), Dengzhan Shengmai (12), Qizhitongluo (13) and NeuroAiD (14, 15). These trials differ in more than formulation. They enroll patients at different points after stroke, from within 36 h of onset to as long as 6 months afterward, and measure different outcomes over follow-up periods ranging from 2 weeks to 2 years. Previous reviews have noted that the principal limitation of this literature is methodological rather than quantitative (16).

Two consequences follow. Evidence about one formulation cannot be extrapolated to traditional Chinese medicine as a whole, and evidence generated in the acute phase cannot be extrapolated to rehabilitation or secondary prevention. When trials differ this widely in population, intervention, comparator, outcome, and follow-up, combining them into a single summary estimate yields a number that no clinician or guideline developer could act on. Synthesis Without Meta-analysis (SWiM) guidance was developed for this situation (17), and GRADE allows certainty to be rated transparently for each outcome whether or not pooling is possible (18).

We therefore conducted a systematic review reported with reference to PRISMA 2020 (19) and PRISMA-S (20), with structured narrative synthesis under SWiM, risk-of-bias assessment using RoB 2 (21), and GRADE certainty rating. Our objectives were to characterize the contemporary randomized evidence, to present each trial using the effect measure it published, to organize the synthesis by clinical phase and formulation, and to summarize certainty of evidence and methodological priorities for future trials. We did not undertake a meta-analysis, and we make no claim about traditional Chinese medicine as a unified treatment class.

Methods

Protocol, registration and reporting

This review was not prospectively registered in PROSPERO or any other prospective register, and no dated written protocol or statistical analysis plan was completed before screening began. The consequence is specific: readers cannot verify independently that the eligibility criteria, outcomes of interest and analytic approach were fixed before the results of the individual trials were known, and *post hoc* modification of those decisions or selective emphasis among outcomes cannot be excluded. The review has not been registered retrospectively, because retrospective registration would not remedy this limitation and could be mistaken for prospective registration. Accordingly, no analysis reported here is described as pre-specified and no decision is described as having been made in advance. PRISMA 2020 (19) and PRISMA-S (20) were used as reporting frameworks. Items that could not be completed because the original records were unavailable are identified explicitly in the manuscript and [Supplementary material](#). The narrative synthesis follows the SWiM reporting guideline (17). GRADE was used to rate certainty by outcome, formulation and clinical phase (18).

Eligibility criteria (PICOS)

Population

Adults with ischemic stroke confirmed by computed tomography or magnetic resonance imaging. Trials of intracerebral hemorrhage, subarachnoid hemorrhage or transient ischemic attack without imaging-confirmed infarction were not eligible. Trials enrolling mixed stroke populations were eligible only if ischemic subgroup results were reported separately.

Intervention

Any traditional Chinese medicine formulation given in addition to conventional stroke care, including single herbs, classical compound prescriptions, Chinese patent medicines and proprietary multi-component preparations. Oral, nasogastric and intravenous routes were eligible. Acupuncture, moxibustion, tuina and other non-pharmacological TCM modalities were not eligible, because the review question concerns pharmacological adjuncts.

Comparator

Placebo plus standard care, or standard care alone. Standard care means guideline-concordant management of acute ischemic stroke or of secondary prevention as defined by the guidelines in force at each trial site and time, comprising antiplatelet or anticoagulant therapy as indicated, lipid-lowering therapy, management of blood pressure and glucose, and reperfusion therapy where eligible and available. Trials in which both arms received a traditional Chinese medicine preparation, with no